
$\mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash \mathcal{BRE}, G \Vdash A$ identity
$\mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash \mathcal{ERG}, G \Vdash A$ identity
$\mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash (\mathcal{BRE}) \wedge (\mathcal{ERG}), G \Vdash A$ andR
$\mathcal{BRG}, \mathcal{ERG}, \mathcal{BRE} \vdash \mathcal{BRG}, G \Vdash A$ identity
$G \Vdash A, \mathcal{BRG}, \mathcal{ERG}, \mathcal{BRE} \vdash G \Vdash A$ identity
$(\mathcal{BRG}) \to (G \Vdash A), \mathcal{BRG}, \mathcal{ERG}, \mathcal{BRE} \vdash G \Vdash A$ implyL
$\forall D.((\mathcal{BR}D) \to (D \Vdash A)), \mathcal{BRG}, \mathcal{ERG}, \mathcal{BRE} \vdash G \Vdash A$ forallL
$\mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)), \mathcal{BRG}, \mathcal{ERG} \vdash G \Vdash A$ shuffleL
$\mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)), \mathcal{BRG} \vdash G \Vdash A$ shuffleL
$\mathcal{BRG}, \mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash G \Vdash A$ shuffleL
$((\mathcal{BRE}) \wedge (\mathcal{ERG})) \to (\mathcal{BRG}), \mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash G \Vdash A$ implyL
$\forall u.(((\mathcal{BRE}) \wedge (\mathcal{ER}u)) \to (\mathcal{BR}u)), \mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash G \Vdash A$ forallL
$\forall v.(\forall u.(((\mathcal{BR}v) \wedge (v\mathcal{R}u)) \to (\mathcal{BR}u))), \mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash G \Vdash A$ forallL
$\forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))), \mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)) \vdash G \Vdash A$ forallL
$\forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))), \mathcal{ERG}, \mathcal{BRE} \vdash G \Vdash A$ shuffleL
$\mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))), \mathcal{ERG} \vdash G \Vdash A$ shuffleL
$\mathcal{ERG}, \mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash G \Vdash A$ shuffleL
$\mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash (\mathcal{ERG}) \to (G \Vdash A)$ implyR
$\mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash \forall F.((\mathcal{ER}F) \to (F \Vdash A))$ forallR
$\mathcal{BRE}, \forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash E \Vdash \Box A$ mBoxR
$\forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash (\mathcal{BRE}) \to (E \Vdash \Box A)$ implyR
$\forall D.((\mathcal{BR}D) \to (D \Vdash A)), \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash \forall C.((\mathcal{BR}C) \to (C \Vdash \Box A))$ forallR
$B \Vdash \Box A, \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash \forall C.((\mathcal{BR}C) \to (C \Vdash \Box A))$ mBoxL
$B \Vdash \Box A, \forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash B \Vdash \Box(\Box A)$ mBoxR
$\forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash (B \Vdash \Box A) \to (B \Vdash \Box(\Box A))$ implyR
$\forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash B \Vdash (\Box A) \to (\Box(\Box A))$ mImplyR
$\forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u)))) \vdash \forall t.(t \Vdash (\Box A) \to (\Box(\Box A)))$ forallR
$\vdash (\forall w.(\forall v.(\forall u.(((w\mathcal{R}v) \wedge (v\mathcal{R}u)) \to (w\mathcal{R}u))))) \to (\forall t.(t \Vdash (\Box A) \to (\Box(\Box A))))$ implyR